

TAARA Q.0

Quantum Behavioral Analysis Platform

User Manual — v1.0

GoodWinSun · Confidential

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TAARA detects behavioral anomalies that classical tools miss — not by setting thresholds, but by measuring the geometric distance between current system state and every prior normal state using quantum fidelity.

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1. What is TAARA?

TAARA (Threat-Aware Adaptive Reasoning Agent) is a cybersecurity platform for independent security consultants and managed service providers. It monitors servers, cloud infrastructure, and application environments by analyzing behavioral patterns — not known signatures.

Most security tools work by matching activity against a list of known attacks. If the attack is new, unknown, or deliberately crafted to avoid signatures, it goes undetected. TAARA takes a fundamentally different approach: it learns what *normal* looks like for a specific system and measures how far the current behavior has drifted from that baseline.

The measurement is quantum-geometric: each behavioral snapshot is encoded as a quantum state and compared to all prior normal states using **fidelity** — a cosine-like inner product in Hilbert space. A result below 0.5 means the current state is more orthogonal (dissimilar) than parallel to anything seen before. No threshold to tune. No signatures to update. No false-positive calibration.

PRIMARY USE CASES

SSH server monitoring	Detects lateral movement, privilege escalation, login anomalies
Cloud workload analysis	Monitors AWS/GCP/Azure instance behavioral drift
Post-incident review	Identifies the moment of behavioral change with F_min timestamp
Autonomous remediation	Agent proposes and executes containment with operator approval

2. First Connection

TAARA connects to servers over SSH. No agent is required on the target server for initial analysis — TAARA reads logs, process lists, and network state remotely over the SSH session.

STEP-BY-STEP

1 Open the TAARA app

Click the TAARA icon on your desktop or run `launch_taara.sh`.

2 Select a client or add new

The home screen shows your client portfolio. Click a client card to pre-fill connection details, or click + Add Client.

3 Enter SSH credentials

Host IP/hostname, port (default 22), username, and either password or key path.

4 Click Connect →

TAARA establishes the SSH session and begins collecting the 17-feature behavioral vector.

5 Run TAARA Analysis

Click the blue Run Analysis button. TAARA collects logs, checks auth events, and runs quantum fidelity scoring. This takes 30-60 seconds.

6 Review results

The analysis dashboard shows findings, F_min score, MITRE ATT&CK; mappings, and recommended fixes.

Note: If you don't have a live server to connect to, use **Demo Mode** (button on the Client Portfolio screen). Demo Mode runs a synthetic SSH intrusion scenario with real quantum math — all F_min values and findings are computed from the actual quantum engine, not hardcoded.

3. Understanding F_min

F_min is TAARA's core output. Every analysis produces a single number between 0 and 1. It is not a score, a percentage, or a risk rating — it is a mathematical quantity from quantum information theory.

$$F = |\langle \psi_t | \psi_m \rangle|^2$$

where ψ_t is the quantum state encoding current behavior and ψ_m is the closest state in the behavioral memory basis

What the formula means in plain language:

TAARA converts the 17-feature behavioral vector into a unit vector in a high-dimensional Hilbert space using angle encoding. Each prior normal state is also stored as a unit vector. F_min is the squared inner product between the current state and the most similar prior normal state.

A value of **1.0** means current behavior is identical to something seen before — completely normal. A value of **0.0** means current behavior is completely orthogonal to every prior normal state — geometrically as anomalous as possible. The threshold of **0.5** is not arbitrary: it is the midpoint of Hilbert space, the point at which a state is equally similar and dissimilar to the memory basis.

F_MIN INTERPRETATION

0.7 – 1.0	NORMAL	Behavior matches prior normal states. No action required.
0.5 – 0.7	DRIFTING	Behavioral drift detected. Monitor closely. May indicate misconfiguration or early-stage anomaly.
0.3 – 0.5	UNSAFE DIRECTION	Current state is more orthogonal than parallel to normal basis. Investigation recommended.
0.0 – 0.3	CRITICAL DIVERGENCE	Extreme behavioral deviation. Quantum-confirmed anomaly. Immediate action required.

4. Reading a Finding

Each finding in the analysis results represents a specific security issue detected on the monitored system. Findings include a severity, a description, a quantum deviation score, and an actionable fix command.

FINDING ANATOMY

Severity	CRITICAL / HIGH / MEDIUM / LOW / INFO — how urgently this needs to be addressed.
CVE / ID	Identifier for the vulnerability or finding type. CVE IDs link to NVD.
Description	What was found and why it matters in plain language.
F_dev score	F_dev = deviation of this specific feature from its safe-state quantum embedding. High F_dev means this feature was the primary driver of the anomaly.
Fix command	Exact shell command to remediate, ready to paste into a terminal.
MITRE tactic	The ATT&CK; tactic this finding maps to — useful for client reports and compliance.

ANOMALY ALERT BANNER

When F_min drops below 0.5 during live monitoring or analysis, TAARA fires a full-width red anomaly banner at the top of the screen. The banner shows: server name, F_min value, top offending features, correlation detection status, and the number of agent actions auto-executed or awaiting approval. This banner cannot be missed — it is the visual confirmation that TAARA has caught something.

5. Agent Autonomy Levels

TAARA's agent component (TaaraWare) can take remediation actions automatically or propose them for operator approval. The autonomy level controls this behavior.

0%	Observe only Agent detects anomalies and logs them. No actions proposed or taken. Use for initial assessment periods.
25%	Suggest only Agent proposes actions and explains rationale. All actions require manual approval in the TaaraWare → Agent & Actions tab.
50%	Supervised Actions with high historical approval rates (>90%) and success rates (>85%) are pre-approved. Novel actions still require approval.
75%	High autonomy Agent executes most actions automatically. Only actions with potential for service disruption are held for approval.
100%	Full autonomy Agent acts immediately on any detected anomaly. Full execution log available in Rollback Log tab. One-click rollback on any action.

CONTRASTIVE BANDIT LEARNING

The autonomy level is not a static setting — it is informed by TAARA's contrastive bandit. The bandit tracks each action type's approval rate (how often operators approve it) and success rate (how often the issue resolved after execution). Actions that consistently get approved and work are promoted to auto-execution. This means TAARA becomes more useful over time as it learns each operator's judgment.

6. Generating a Client Report

After completing an analysis, TAARA can generate a professional PDF report for your client. The report includes: executive summary, all findings with severity and fix commands, quantum fidelity chart, agent action log, and a risk score summary.

TO GENERATE A REPORT

1. Complete a full analysis (Connect → Run Analysis → wait for results).
2. In the Analysis view, click **Generate Report** in the top-right of the results panel.
3. TAARA generates the PDF using the LLM-powered executive summary engine.
4. The report opens automatically in your system PDF viewer.
5. Send the PDF directly to your client — it is formatted for non-technical readers.

REPORT PRICING

TAARA enables independent consultants to offer paid security assessments at ■15,000 per engagement. The report is the deliverable — a professional document the client receives after TAARA runs the analysis. No manual writeup required.

7. Glossary

Quantum Fidelity

A measure of how similar two quantum states are. Defined as $F = |\langle \psi | \phi \rangle|^2$. Value of 1 = identical states, 0 = orthogonal (maximally dissimilar).

F_min

The minimum fidelity between the current behavioral state and any state in the normal memory basis. TAARA's primary anomaly signal. Below 0.5 triggers an alert.

Angle Encoding

A quantum circuit technique where feature values are encoded as rotation angles in qubit states. TAARA uses AngleEmbedding with parameterized RX/RZ/RY gates followed by Ring-CNOT entanglement layers.

Behavioral Memory

The set of quantum states representing all prior normal observations for a given identity. TAARA builds this memory during the baseline period and updates it during operation.

Correlation Signal

Detected when $F_{\text{angle}} < F_{\text{amplitude}} - 0.05$. Indicates the directional (angle-encoded) representation shows a stronger anomaly than the magnitude-only representation — signature of coordinated multi-feature attacks.

Contrastive Bandit

TAARA's reinforcement learning component that tracks action outcomes and adjusts auto-execution thresholds. Uses approval_rate and success_rate per action type.

PQC Kyber768

Post-quantum cryptographic key encapsulation mechanism standardized as ML-KEM (NIST FIPS 203). Protects the TaaraWare $\leftrightarrow$ CommandCenter channel against Shor's algorithm attacks on RSA/ECC.

TaaraWare

The lightweight agent deployed to the monitored server. Collects the 17-feature behavioral vector every 10 seconds, runs local quantum fidelity checks, and reports back over the PQC channel.

17-Feature Vector

The behavioral fingerprint TAARA collects: CPU, memory, disk, network I/O, connections, process count, load averages (1/5/15m), failed logins, new processes, suspicious connections, privilege escalations, temporal rhythm deviation, causal chain novelty, and concealment signal.

Harvest-Now-Decrypt-Later

An attack strategy where adversaries capture encrypted traffic today, storing it to decrypt once quantum computers break current cryptography. PQC Kyber768 is the defense against this.

8. Benchmark Methodology

TAARA's benchmark claims are based on controlled evaluation against public intrusion detection datasets using a held-out test set. This section documents the methodology so results can be independently reproduced.

DATASETS USED

SSH Log Dataset	Structured SSH authentication logs with labeled brute-force and intrusion events.
CICIDS 2017	Canadian Institute for Cybersecurity Intrusion Detection Evaluation Dataset 2017.
Synthetic Demo (internal)	Generated using fixed seed (<code>np.random.default_rng(42)</code>) for reproducibility. Not used in published benchmarks.

EVALUATION PROTOCOL

Models are trained on the first 70% of temporal data (preserving time order to avoid look-ahead bias). Evaluation is on the remaining 30%. Metrics: F1-score, precision, recall, and false positive rate at the same operating point. TAARA's quantum fidelity detector is compared against: Isolation Forest (baseline), One-Class SVM, and Local Outlier Factor.

HONESTY STATEMENT

TAARA's quantum circuit runs on classical simulation (PennyLane `default.qubit`). No quantum speedup is claimed. The advantage over classical methods is geometric: angle encoding detects directional anomalies that amplitude-only methods miss, specifically correlated multi-feature attacks where each feature individually appears within normal range but the combination is anomalous.